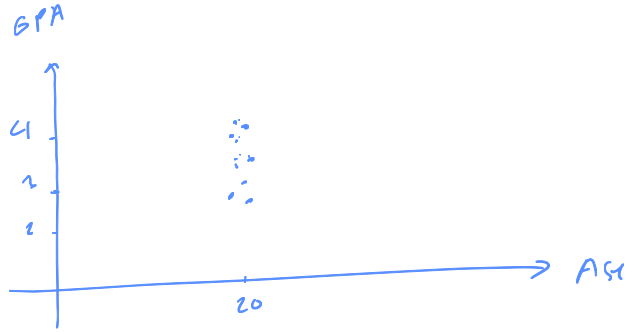
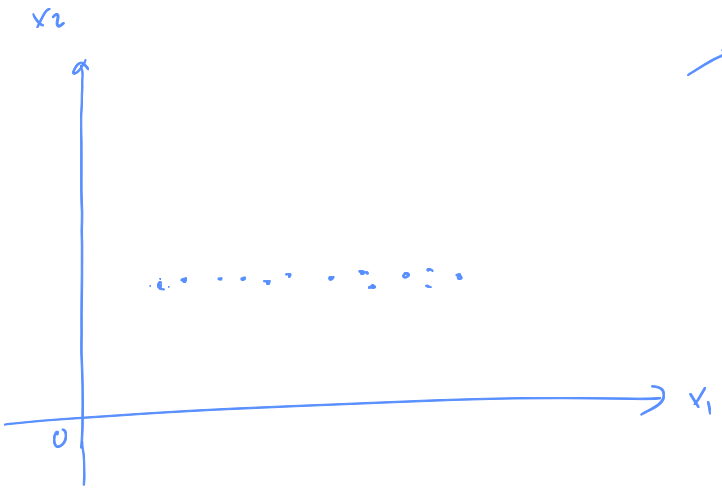
Age	GPA
20	3.5
20	2.1
20	4.0
20.5	3.9
20	1.6
20	2.9
19.5	3.5

① GPA contains more information than Age.

② Age is almost a constant, does not contain much information.

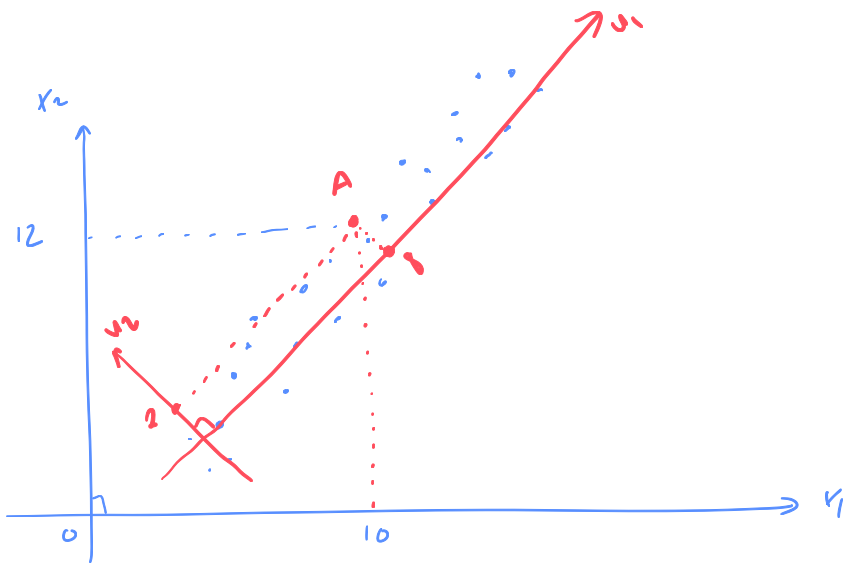
③ If we want to remove a variable, we would remove "Age"





① X_2 does not contain much information

② If we want to remove a variable, we would remove X_2

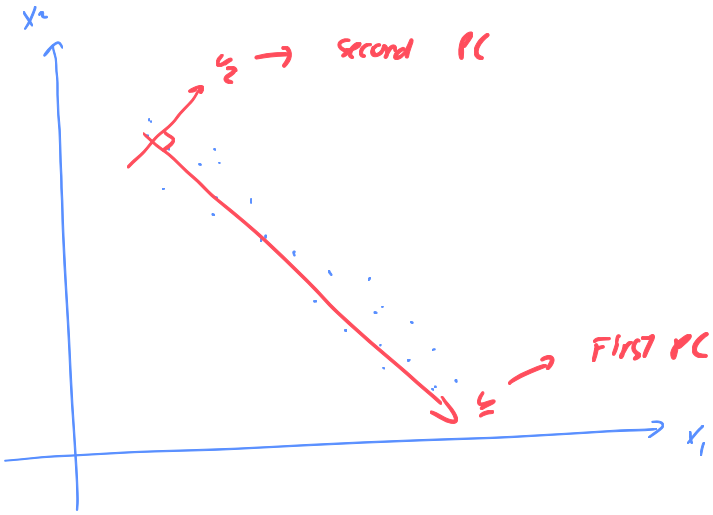


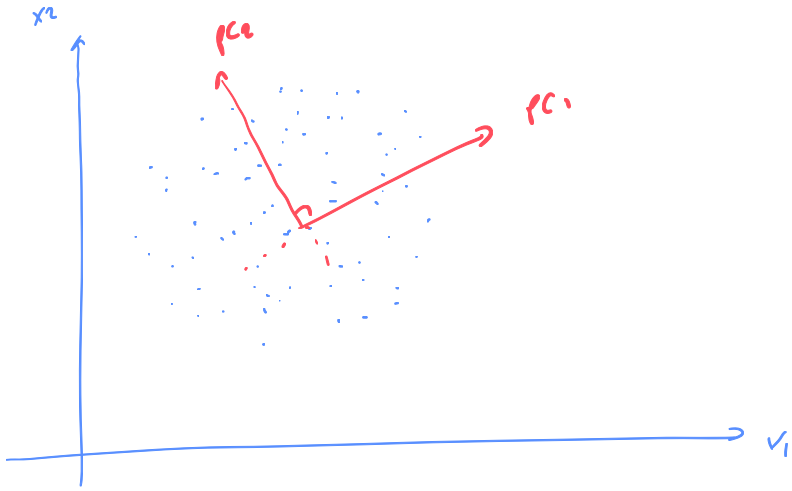
	X_1	X_2		U_1	U_2
(A)	10	12	→	8	2

U_1: The first principal component

U_2: The second principal component.

we can remove U_2 if we want the data has only 1 variable.





In this case, PCA does not really help.